

Gauge fixing for GFFA all left-invariant basis

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Abstract

Because the usual HMC relies on all left-invariant differential operators, it is simpler to just be consistent and calculate the entire Jacobian in terms of left-invariant basis. This calculation is done here.

1 Gauge-fixing flow step

Left-invariant Jacobian

$$\mathcal{F}^{ab}(z, \rho; y, \nu) = -2 \text{Tr} \left\{ \partial_{z, \rho}^b U(y, \nu) U^\dagger(y, \nu) T^a \right\}. \quad (1)$$

Single gauge-fixing step at lattice site x :

$$[\mathcal{E}(x)](y, \nu) = \begin{cases} e^{-t\Omega(x)} U(y, \nu), & y = x \\ U(y, \nu) e^{+t\Omega(x)}, & y = x - \hat{\nu}, 1 \leq \nu \leq 4 \\ U(y, \nu), & \text{else.} \end{cases} \quad (2)$$

Consider the following

$$\begin{aligned} d(LgR)(LgR)^{-1} &= (dL \cdot gR + L \cdot dg \cdot R + Lg \cdot dR)(R^{-1}g^{-1}L^{-1}) \\ &= dLL^{-1} + Ldgg^{-1}L^{-1} + LgdRR^{-1}g^{-1}L^{-1}. \end{aligned} \quad (3)$$

And appropriately choose L and R for the above cases.

1.1 Left-updated term

This is the same as from Ref. [1]. Since I am just considering the left update here, not those other links that are not updated, it looks a bit different than in that ref. The explicit expression is

$$\left[\mathcal{F}(x) \right]_L(y, \nu; z, \rho)^{ac} = \delta_{xy} \left\{ \left(e^{\text{Ad } X} \right)^{ac} \delta_{yz} \delta_{\nu\rho} - tJ(-X)^{ab} \left[O_*(U) \right](y; z, \rho)^{bc} \right\}_{X=\tau\Omega(x)}, \quad (4)$$

with

$$\left[O_*(U) \right](y; z, \rho)^{bc} = \partial_{z, \rho}^c \Omega^b(y). \quad (5)$$

This is calculated explicitly in Section 1.3. For ease of calculation in the code consider factoring out J above. This leaves

$$\left[J(-X) \right]^{-1} \cdot e^{\text{Adj } X} = \left[e^{\text{Adj } -X} J(-X) \right]^{-1} = \left[J(X) \right]^{-1}, \quad (6)$$

where we've used

$$J(X)^T = J(-X) = e^{\text{Adj} X} J(X). \quad (7)$$

The above can therefore be written

$$\left[\mathcal{F}(x) \right]_L (y, \nu; z, \rho)^{ac} = \delta_{xy} J(-X)^{ab} \left\{ \left[J(X)^{-1} \right]^{bc} \delta_{yz} \delta_{\nu\rho} - t \left[O_*(U) \right] (y, \nu; z, \rho)^{bc} \right\}_{X=\tau\Omega(x)}, \quad (8)$$

1.2 Right-updated term

From Eq. (3)

$$d(gR)(gR)^{-1} = dgg^{-1} + gdRR^{-1}g^{-1}, \quad (9)$$

For us

$$\left[\mathcal{F}(x) \right]_R (z, \rho; y, \nu)^{ab} = \delta^{ab} \delta_{y, x-\hat{\nu}} \delta_{zy} \delta_{\rho\nu} - 2\delta_{y, x-\hat{\nu}} \text{Tr} \left\{ U(y, \nu) \partial_{z, \rho}^b e^{t\Omega(x)} \cdot e^{-t\Omega(x)} U^\dagger(y, \nu) T^a \right\} \quad (10)$$

With Lüscher's notation

$$\partial_{x, \mu}^a e^X e^{-X} = J(-X) \cdot X_*. \quad (11)$$

Therefore

$$\begin{aligned} \left[\mathcal{F}(x) \right]_R (z, \rho; y, \nu)^{ab} &= \delta_{y, x-\hat{\nu}} \left[\delta^{ab} \delta_{zy} \delta_{\rho\nu} - 2 \text{Tr} \left\{ U^\dagger(y, \nu) T^a U(y, \nu) \left[-tJ(X) \right]^{bf} \left[O_*(U) \right] (y; z, \rho)^{fd} T^d \right\}_{X=\tau\Omega} \right] \\ &= \delta_{y, x-\hat{\nu}} \left[\delta^{ab} \delta_{y, x-\hat{\nu}} \delta_{zy} \delta_{\rho\nu} + 2tJ(X)^{bf} \left[O_*(U) \right] (y; z, \rho)^{fd} \text{Tr} \left\{ U^\dagger(y, \nu) T^a U(y, \nu) T^d \right\}_{X=\tau\Omega} \right]. \end{aligned} \quad (12)$$

Consider

$$U^\dagger T^a U = \left(\text{Adj}_{U^{-1}} \right)^{ab} T^b, \quad (13)$$

Therefore

$$-2 \text{Tr} \left\{ U^\dagger(y, \nu) T^a U(y, \nu) T^d \right\} = \left(\text{Adj}_{U^{-1}} \right)^{ak} \delta^{kd}. \quad (14)$$

We can also consider the fact that the adjoint representation of the unitary group is a subgroup of a larger orthogonal group so that $[\text{Adj}_{U^{-1}}]^T = \text{Adj}_U$ Using in the above we have ([check sign on tJ](#))

$$\left[\mathcal{F}(x) \right]_R (z, \rho; y, \nu)^{ab} = \delta_{y, x-\hat{\nu}} \left[\delta^{ab} \delta_{zy} \delta_{\rho\nu} - tJ(X)^{bf} \left[O_*(U) \right] (y, \nu; z, \rho)^{fd} \left(\text{Adj}_U \right)^{da} \right]_{X=\tau\Omega}. \quad (15)$$

1.3 Derivative of flow generator

We need to calculate

$$\left[O_*(U) \right] (y, \nu; z, \rho)^{bc} = \partial_{z, \rho}^c \Omega^b(y). \quad (16)$$

Consider that Ω is a projection into the Lie algebra $\mathfrak{su}(3)$:

$$\Omega(x) = \mathcal{P} \left(\sum_{\mu} \left[U(x, \mu) - U(x - \hat{\mu}, \mu)^\dagger \right] \right). \quad (17)$$

Since $X, \Omega \in \mathfrak{su}(3)$, $f : \mathfrak{su}(3) \rightarrow \mathfrak{su}(3)$.

Consider

$$\begin{aligned}\partial_{z,\rho}^c U(x, \mu) &= T^c U(x, \mu) \delta_{x,z} \delta_{\rho,\mu} \\ \partial_{z,\rho}^c U^\dagger(x, \mu) &= -U^\dagger(x, \mu) T^c \delta_{x,z} \delta_{\rho,\mu}.\end{aligned}\tag{18}$$

Therefore we have

$$\partial_{z,\rho}^c \Omega(y) = \mathcal{P}\left(T^c U(y, \nu)\right) \delta_{x,y} \delta_{\rho,\nu} + \mathcal{P}\left(U^\dagger(y - \hat{\nu}, \nu) T^c\right) \delta_{z,y-\hat{\nu}} \delta_{\rho,\nu}.\tag{19}$$

To extract the second index consider

$$\mathcal{P}(M) = \text{Re Tr} \left(T^a M \right) T^a.\tag{20}$$

1.4 Total Jacobian

$$\begin{aligned}\left[\mathcal{F}(x)\right](z, \rho; y, \nu)^{ab} &= (\text{unupdated links}) \delta_{yz} \delta_{\nu\rho} \delta^{ab} \\ &+ \delta_{xy} J(-X)^{ab} \left\{ \left[J(X)^{-1} \right]^{bc} \delta_{yz} \delta_{\nu\rho} - t \left[O_*(U) \right] (y, \nu; z, \rho)^{bc} \right\}_{X=\tau\Omega(x)} \\ &+ \delta_{y,x-\hat{\nu}} \left[\delta^{ab} \delta_{zy} \delta_{\rho\nu} - t J(X)^{bf} \left[O_*(U) \right] (y, \nu; z, \rho)^{fd} \left(\text{Adj}_U \right)^{da} \right]_{X=\tau\Omega}.\end{aligned}\tag{21}$$

2 Derivatives of log det Jacobian

I don't want to take the derivative of inverses of matrices so I'll start from the Jacobian

$$\begin{aligned}\left[\mathcal{F}(x)\right](y, \nu; z, \rho)^{ac} &= \delta_{xy} \left\{ \left(e^{\text{Ad } X} \right)^{ac} \delta_{yz} \delta_{\nu\rho} - t J(-X)^{ab} \left[O_*(U) \right] (y, \nu; z, \rho)^{bc} \right\}_{X=\tau\Omega(x)} \\ &+ \delta_{y,x-\hat{\nu}} \left[\delta^{ac} \delta_{zy} \delta_{\rho\nu} - t J(X)^{bf} \left[O_*(U) \right] (y, \nu; z, \rho)^{fd} \left(\text{Adj}_U \right)^{da} \right]_{X=\tau\Omega}.\end{aligned}\tag{22}$$

consider

$$d \ln \det F = \text{tr} \left(F^{-1} dF \right).\tag{23}$$

In our case we want

$$\partial_{n,\kappa}^d \ln \det \left[\mathcal{F}(x) \right] (y, \nu; z, \rho)^{ac}.\tag{24}$$

2.1 Numerical evaluation

It may be easier to implement this portion with numerical rather than analytic derivatives. Hopefully this doesn't degrade the algorithm appreciably?

$$\partial_{z,\rho}^a f(U) = \lim_{\epsilon \rightarrow 0} \frac{f(e^{\epsilon T^a} U) - f(U)}{\epsilon}\tag{25}$$

A Derivative of adjoint action

We'll need to evaluate

$$\partial_{z,\rho}^c \left(\text{Adj}_{U^{-1}} \right)^{ad}. \quad (26)$$

Consider

$$\begin{aligned} \partial_{z,\rho}^c \text{Tr} \left\{ U^\dagger(y, \nu) T^a U(y, \nu) T^d \right\} &= \delta_{zy} \delta_{\rho, \nu} \text{Tr} \left\{ \left[T^c U(y, \nu) \right]^\dagger T^a U(y, \nu) T^d + U^\dagger(y, \mu) T^a T^c U(y, \nu) T^b \right\} \\ &= \delta_{zy} \delta_{\rho, \nu} \text{Tr} \left\{ U^\dagger(y, \mu) T^a T^c U(y, \nu) T^b - U(y, \nu)^\dagger T^c T^a U(y, \nu) T^d \right\} \\ &= \delta_{zy} \delta_{\rho, \nu} \text{Tr} \left\{ U^\dagger T^a U U^\dagger T^c U T^b - U^\dagger T^c U U^\dagger T^a U T^d \right\} \\ &= \delta_{zy} \delta_{\rho, \nu} \left[\text{Adj}_{U^{-1}} \right]^{af} \left[\text{Adj}_{U^{-1}} \right]^{cg} \text{Tr} \left\{ T^b [T^f, T^c] \right\} \end{aligned} \quad (27)$$

Using $[T^a, T^b] = f^{abc} T^c$

$$\begin{aligned} \partial_{z,\rho}^c \left(\text{Adj}_{U^{-1}} \right)^{ad} &= \delta_{zy} \delta_{\rho, \nu} \left[\text{Adj}_{U^{-1}} \right]^{af} \left[\text{Adj}_{U^{-1}} \right]^{cg} f^{fce} \left(-\frac{1}{2} \delta^{be} \right) \\ &= \delta_{zy} \delta_{\rho, \nu} \left[\text{Adj}_{U^{-1}} \right]^{af} \left[\text{Adj}_{U^{-1}} \right]^{cg} f^{fcb} \end{aligned} \quad (28)$$

And I should have done this way originally but now that it is written out this way it is obvious that

$$\partial_{z,\rho}^c \left(\text{Adj}_U \right)^{ad} = \delta_{zy} \delta_{\rho, \nu} \left[\text{Adj}_U \right]^{af} \left[\text{Adj}_U \right]^{cg} f^{fcb} \quad (29)$$

B Revision history

- 05/08/25: Began and Jacobians worked out in full.
- TODO: Add derivatives of log det Jacobian.

References

- [1] Martin Lüscher. “Trivializing maps, the Wilson flow and the HMC algorithm”. In: *Commun. Math. Phys.* 293 (2010), pp. 899–919. DOI: 10.1007/s00220-009-0953-7. arXiv: 0907.5491 [hep-lat].